

The *for* Loop

Section 4

Chapter 3

An Introduction to Programming Using Python
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Loops

- Topics
 - For loops
 - Range function
 - Nested loops

The *for* Loop

- Used to iterate through a sequence of values
- General form of a for loop
- Sequence can be
 - Arithmetic progression of numbers
 - String
 - List
 - File object

```
for var in sequence:  
    indented block of statements
```

The *for* Loop

- Variable is successively assigned each value in the sequence

```
for var in sequence:  
    indented block of statements
```

- Indented block of statements executed after each assignment
- Physical indentation tells interpreter where block starts and stops

For Loop Example

- Example:

```
fruits = ['apple', 'pear', 'plum', 'peach']  
for item in fruits:  
    print(item)
```

Output:

```
apple  
pear  
plum  
peach
```

Your Turn ...

```
for num in [1, 4, 8, 3, 11, 25, 9, 16]:  
    if num > 20:  
        print('Exiting loop')  
        break  
    elif num%2 == 0:  
        continue  
    print(num)
```

Output:

1

3

11

Exiting loop

Looping Through Arithmetic Progression of Numbers

- Range function is used to generate an arithmetic progression

`range(3, 10)` generates the sequence 3, 4, 5, 6, 7, 8, 9.

`range(0, 4)` generates the sequence 0, 1, 2, 3.

`range(-4, 2)` generates the sequence -4, -3, -2, -1, 0, 1.

range(n)

– *range(n)*: produces **sequence** 0, 1, 2, ... $n-1$

```
L = [1, 5, 7, 2, 8]
```

```
for i in range(len(L)):
```

```
    L[i] = L[i]+10
```

```
print(L)
```

- [11, 15, 17, 12, 18]

Step Values for the *range* Function

- Variation of the range function generates a sequence of integers
 - **range([*i*,]stop[, *k*])**: sequence starts at *i* (instead of 0) and incremented by *k* (instead of 1)
 - **range(3, 20, 4)** → produces 3, 7, 11, 15, 19
- Examples: sequence generated by

range (3, 10, 2) →	3, 5, 7, 9
range(0,24, 5) →	0, 5, 10, 15, 20
range (-10, 10, 4) →	-10, -6, -2, 2, 6
range(5, 10) →	5, 6, 7, 8, 9

Step Values for the *range* Function

- If negative step value is used and initial value is greater than terminating value,
 - Range function generates a decreasing sequence
- Examples:

`range(6, 0, -1)` generates the sequence 6, 5, 4, 3, 2, 1.

`range(5, 2, -3)` generates the sequence 5.

`range(10, -10, -4)` generates the sequence 10, 6, 2, -2, -6.

Looping Through Arithmetic Progression of Numbers

- Example 1: Code displays four integers and their squares

```
for i in range(2, 6):  
    print(i, i * i)
```

[Run]

```
2 4  
3 9  
4 16  
5 25
```

Nested For/While Loops

- The block of code under an ***for*** loop or ***while*** loop may contain any type of statements, including other loops.
 - These are called ***nested*** loops
 - These must be indented further to indicate which statement they are under.
 - There may be multiple levels of nesting.
 - Amount of indentation indicates which statement a suite of code belongs to.

Nested *for* Loops

- Body of for loop can contain another for loop.
- Second loop must be completely contained inside the first loop
 - Must have a different loop variable

Nested *for* Loops

- Example 5: Program uses nested for loops to display a triangle of asterisks.

```
## Display a triangle of asterisks.  
numberOfRows = int(input("Enter a number from 1 through 20: "))  
for i in range(numberOfRows):  
    for j in range(i + 1):  
        print("*", end="")  
    print()
```

[Run]

Enter a number from 1 through 20: 5

*

**

Looping Through Characters of a String

- Example 6: Program requests a word as input and displays it backwards.

```
## Reverse the letters in a word.  
word = input("Enter a word: ")  
reversedWord = ""  
for ch in word:  
    reversedWord = ch + reversedWord  
print("The reversed word is " + reversedWord + ".")
```

[Run]

```
Enter a word: zeus  
The reversed word is suez.
```

Looping Through Items of a List

- Example 8: Program replaces the name of each month with its three-letter abbreviation.

```
## Replace each month with its three-letter abbreviation.  
months = ["January", "February", "March", "April", "May", "June",  
          "July", "August", "September", "October", "November", "December"]  
for i in range(len(months)):  
    months[i] = months[i][0:3]  
print(months)
```

[Run]

```
['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct',  
 'Nov', 'Dec']
```


Your Turn ...

- Example:

```
for i in range(3):  
    for x in 'cat':  
        print (i, ':', x*i)
```

- **Output:**

```
0 :  
0 :  
0 :  
1 : c  
1 : a  
1 : t  
2 : cc  
2 : aa  
2 : tt
```

End

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